

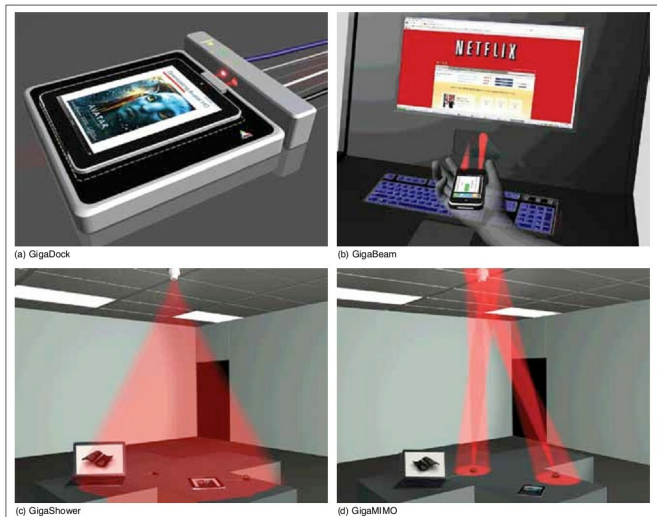


Fig. 2: GigaSpeed usage models (Images courtesy: TriLumina Corp.)

one billion barrels of oil could be saved every year, which again translates into energy production of 250 nuclear power plants.

Driven by the progress of LED technology, visible light communication is gaining attention in research and development. The VLC Consortium (VLCC) in Japan was one of the first to introduce this technology.

After establishing a VLC interest group within the IEEE 802.15 wireless personal-area networks working group, the IEEE 802.15.7 task group was established by the industry, research institutes and universities in 2008. The final standard was approved in 2011. It specifies VLC comprising mobile-to-mobile (M2M), fixed-to-mobile (F2M) and infrastructure-to-mobile (I2M) communications. There, the

focus is on low-speed, medium-range communications for intelligent traffic systems and on high-speed, short-range M2M and F2M communications to exchange, for example, multimedia data. Data rates are supported from some 100 kbps up to 100 Mbps using different modulation schemes.

Other standardisation groups are working on standardised optical wireless communication (OWC) solutions using visible and infrared light. The most important groups are IrDA with its new 10 Giga-IR working group, ISO and ICOSA.

Li-Fi—the superset of VLC & Co.

VLC represents only a fraction of what appears to be a much larger movement towards optical wireless technolo-

gies in general. This larger world has been dubbed 'Li-Fi' (Light Fidelity) by people such as Dr Harald Haas of Edinburgh University and organisations such as the Li-Fi Consortium.

In that connection, Li-Fi comprises several optical wireless technologies such as optical wireless communication, navigation and gesture recognition applied for natural user interfaces (Fig. 1). Thus it provides a completely new set of optical technologies and techniques to offer users add-on as well as complementary functionalities compared to well-known and established RF services. This could reach from a new user experience regarding communication speeds in the gigabit-class to bridge the well-known spectrum crunch, over to precise indoor positioning or controlling video games,